Technology Data Sheet (TDS)

TINMORRY PA6-CF

Version 1.0

• Basic Introduction

TINMORRY PA6-CF is a carbon fiber (CF) reinforced polyamide 6 (nylon 6) composite material. Similar to generic PA6-CF, it has excellent strength, stiffness, heat resistance, oil resistance, durability and other properties, as well as relatively low saturated water absorption and other outstanding properties: significantly higher toughness and layer strength. In addition, after absorbing water and getting damp, its toughness will be further improved, and its resistance to dynamic loads such as impacts, drops, and collisions will be greatly enhanced. In short, TINMORRY PA6-CF is a PA6-CF with high toughness and layer strength.

Note: 1. Since the PA6-CF filament contains carbon fiber, please use a nozzle with a diameter of ≥ 0.4 mm to decrease the risk of nozzle clogging; and please use hard enough nozzles such as those made of hardened steel, and do not use relatively soft nozzles such as those made of brass or stainless steel to slow down the wear of them. 2. Please use an enclosed printer to print it to reduce warping and ensure high enough layer strength. 3. To maintain the high strength, stiffness, and heat resistance of the prints please perform water-proof treatment such as oil immersion, painting spraying, and waxing on the surface as soon as possible after printing, and use them in a dry environment. To further improve the toughness of the prints, soaking the prints in water for about 1 day in advance will work well (but their strength, stiffness, and heat resistance will be decreased).

• Specifications

Subjects	Data
Diameter	1.75 ± 0.03 mm
Net filament weight	1000 g ± 0.5%
Length	340 ± 5 m
Spool material	PC ((Temperature resistance: 120 °C)
Spool size	External Diameter: 198 mm; Inner Diameter: 56 mm; Width: 65 mm

• Recommended Printing Settings

Subjects	Data
Drying settings before printing	80 - 120 °C， 8 - 12 h (Blast drying oven, or filament drying box)
Printing temperature and humidity	≤ 75 ℃, ≤ 20% RH （Put in an air-tight container and sealed with desiccant, or put it in a filament drying box and set the temp to be about 70 °C ）
Storage temperature and humidity	≤ 30 ℃, ≤ 20% RH（Sealed with desiccant）
Compatible support material	Itself or support filament for PA
Compatible printer type	enclosed-frame (recommended)
Compatible nozzle material	Hardened steel
Compatible nozzle size	0.6 (recommended), 0.4, 0.8 mm
Compatible plate type	Textured PEI Plate or other plate
Plate surface preparation	Glue
Nozzle temperature	260 - 290 °C
Bed temperature	80 - 100 °C (depending on the printer and plate types)
Printing speed*	Max: 300 mm/s; Recommended: ≤ 200 mm/s
Chamber temperature	< 80 °C

*When printed with Bambu X1C and P1S, and the nozzle temperature is 290 °C, the max printing speed can reach 300 mm/s.

• Properties

The commonly used physical properties, mechanical properties, chemical properties, printing properties and other properties of TINMORRY PA6-CF have been tested and are shown in the tables below.

• Physical Properties

Subjects	Testing Methods	Data
Density	ISO 1183	1.22 g/cm^3
Saturated water absorption rate	25 °C , 55% RH	1.98%
Melt index	280 °C, 2.16 kg	12.4 ± 1.7 g/ 10 min
Glass transition temperature	DSC, 10 °C/min	53 °C
Crystallization temperature	DSC, 10 °C/min	175 °C
Melting temperature	DSC, 10 °C/min	232 °C
Vicar softening temperature	ISO 306, GB/T 1633	202 °C
Heat deflection temperature 1	ISO 75, 1.82 MPa	175 °C
Heat deflection temperature 2	ISO 75, 0.45 MPa	194 °C

• Mechanical Properties: Dry State

Properties	Testing Methods	Data
Tensile strength (XY)	ISO 527, GB/T 1040	94 ± 5 MPa
Tensile strength (Z)	ISO 527, GB/T 1040	52 ±4 MPa
Young's modulus (XY)	ISO 527, GB/T 1040	4752 ± 328 MPa
Young's modulus (Z)	ISO 527, GB/T 1040	2494 ± 285 MPa
Breaking elongation rate (XY)	ISO 527, GB/T 1040	7.5% ± 1.7%
Breaking elongation rate (Z)	ISO 527, GB/T 1040	2.8% ± 1.1%
Bending strength (XY)	ISO 178, GB/T 9341	138 ± 4 MPa
Bending strength (Z)	ISO 178, GB/T 9341	84 ± 3 MPa
Bending modulus (XY)	ISO 178, GB/T 9341	4083 ± 175 MPa
Bending modulus (Z)	ISO 178, GB/T 9341	2370 ± 146 MPa
Impact strength (XY)	ISO 179, GB/T 1043	54.3 ± 3.5 kJ/m²
Impact strength (Z)	ISO 179, GB/T 1043	16.4 ± 2.6 kJ/m²

• Mechanical Properties: Wet State

Properties	Testing Methods	Data
Tensile strength (XY)	ISO 527, GB/T 1040	45 ± 4 MPa
Tensile strength (Z)	ISO 527, GB/T 1040	29 ± 4 MPa
Young's modulus (XY)	ISO 527, GB/T 1040	1842 ± 278 MPa
Young's modulus (Z)	ISO 527, GB/T 1040	758 ± 147 MPa
Breaking elongation rate (XY)	ISO 527, GB/T 1040	15.3% ± 3.2%
Breaking elongation rate (Z)	ISO 527, GB/T 1040	6.2% ± 1.6%
Bending strength (XY)	ISO 178, GB/T 9341	71 ± 3 MPa
Bending strength (Z)	ISO 178, GB/T 9341	39 ± 3 MPa
Bending modulus (XY)	ISO 178, GB/T 9341	1538 ± 164 MPa
Bending modulus (Z)	ISO 178, GB/T 9341	817 ± 138 MPa
Impact strength (XY)	ISO 179, GB/T 1043	89.9 ± 7.3 kJ/m²
Impact strength (Z)	ISO 179, GB/T 1043	62.4 ± 6.2 kJ/m²

• Chemical and Other Properties

Subjects	Data
Component	PA6, carbon fiber
Skin irritation	May cause allergy in some people, and wearing gloves is suggested
Resistance to water	Yes
Resistance to oil and grease	Resistant to most kinds of daily oil and grease
Resistance to organic solvent	Not resistant to some organic solvents
Resistance to weak acid	Yes
Resistance to strong acid	Not resistant
Resistance to weak alkali	Yes
Resistance to strong alkali	Not resistant
Flammability	Yes

Odor during printing	Pungent
Odor during combustion	Pungent

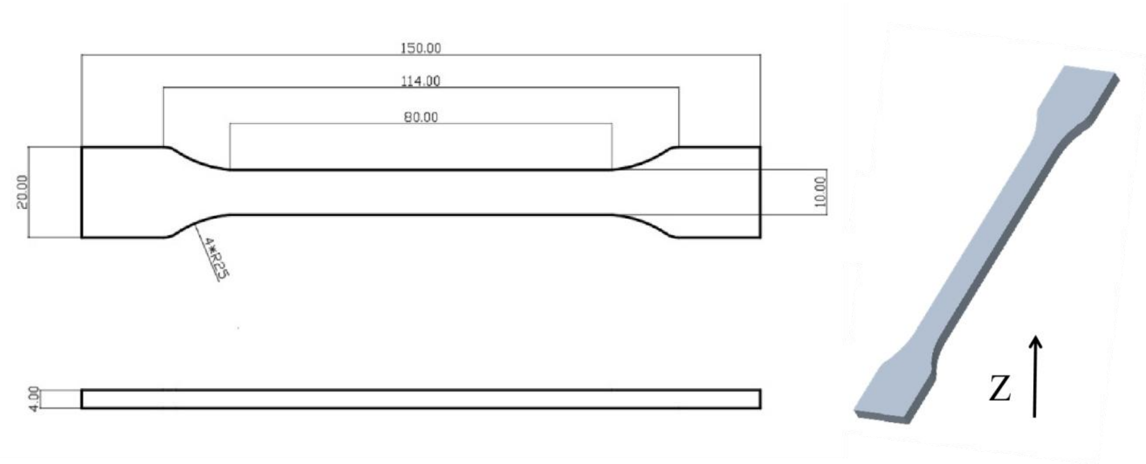
• Specimen Info

Subjects	**Data**
Size	Shown at the following pictures
Nozzle temperature	290 °C
Bed temperature	100 °C
Printing speed	XY: 100 mm/s; Z: 80 mm/s
Infill density	100%
Infill pattern	Concentric

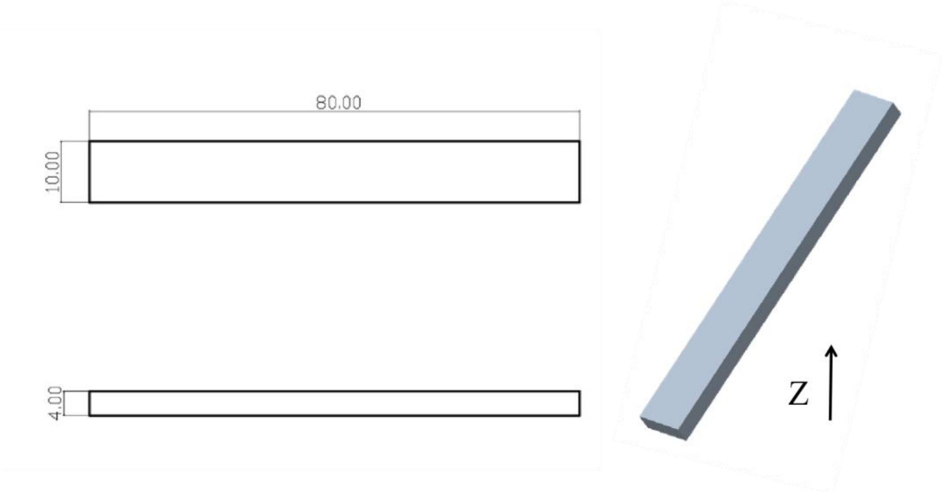
Statements：

1. The test specimens were printed under the key parameters mentioned above. Before testing, all specimens were placed in an indoor environment at about 25 °C for over 12 hours. Please note that when printing, different ambient temperatures, fan speeds and layer times can lead to varying cooling, which can affect the mechanical properties of the specimens, especially those in the Z direction. Additionally, different printers can also result in variations in printing quality.
2. If high printing quality is required, it is recommended to dry every kind of TINMORRY filaments before printing and use an air-tight container and desiccant to protect them from moisture during the printing process (with humidity inside the container less than 20%) to reduce the risk of issues related to moisture absorption in filaments. The recommended drying conditions are shown above. Note that when drying them, avoid placing the spools near the heater to prevent damage to the spools or the filaments due to excessive temperature. Do not use a kitchen oven or microwave oven.
3. After annealing, comprehensive mechanical properties of TINMORRY PA6-CF prints can be slightly increased, while some prints may undergo shrinkage or deformation, so please be aware of this.The recommended annealing process is: first at 90 °C for 4 hours, then slowly heat up to about 120 °C (not recommended to exceed 140 °C) for about 6 hours, and finally slowly cool down in the oven. Prints of different sizes, structures, and shapes can have different annealing performances. If the print is found to be significantly deformed during the process, please switch to a lower annealing temperature.

1. **Tensile Test**



2. **Bending Test and Impact Test**



• Disclaimer

The above properties data is obtained by TINMORRY through testing with standard samples, standard methods, and specific equipment, and is for reference and comparison only; if some data are changed, TINMORRY will update this TDS, but may not be able to inform the users, and please forgive us. The actual performance of 3D prints depends not only on the performance of the filaments used, but also on many factors, such as the degree of moisture absorption of the filaments, the printer, environmental conditions, model characteristics, printing parameters, etc.. When using TINMORRY FDM 3D printing filaments, users are responsible for the legality, safety, and performance of the prints. TINMORRY is responsible for the quality of our filaments, but not for any damage or injury that occurs during using them,or the usage scenarios of them.